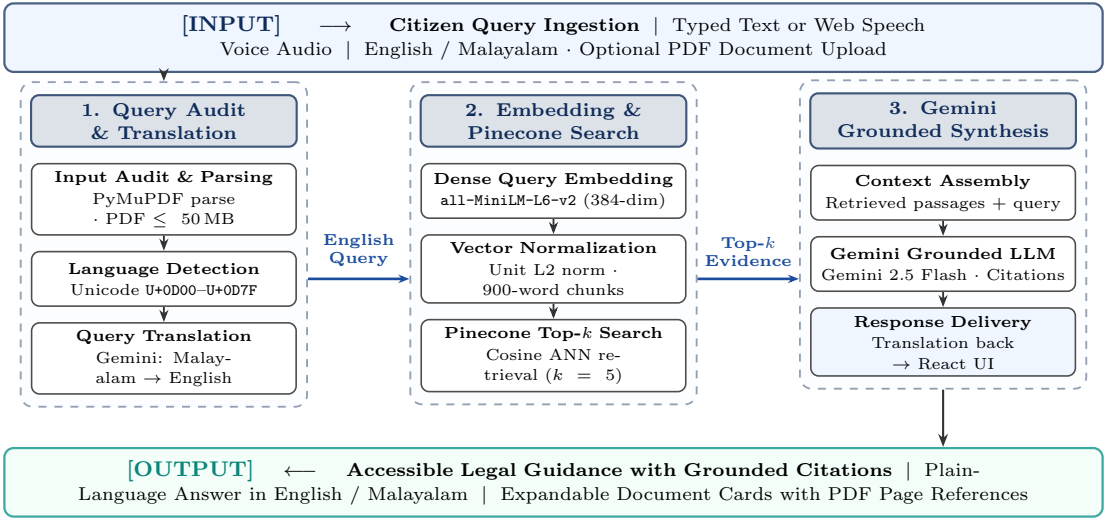

[INPUT] ⟶ Citizen Query Ingestion | Typed Text or Web Speech
Voice Audio | English / Malayalam · Optional PDF Document Upload
1. Query Audit & Translation
Input Audit & Parsing
PyMuPDF parse · PDF ≤ 50 MB
Language Detection
Unicode U+0D00–U+0D7F
Query Translation
Gemini: Malayalam → English
English Query
2. Embedding & Pinecone Search
Dense Query Embedding
all-MiniLM-L6-v2 (384-dim)
Vector Normalization
Unit L2 norm · 900-word chunks
Pinecone Top-k Search
Cosine ANN retrieval (k = 5)
Top-k Evidence
3. Gemini Grounded Synthesis
Context Assembly
Retrieved passages + query
Gemini Grounded LLM
Gemini 2.5 Flash · Citations
Response Delivery
Translation back → React UI
[OUTPUT] ⟵ Accessible Legal Guidance with Grounded Citations | Plain-Language Answer in English / Malayalam | Expandable Document Cards with PDF Page References